New bounds for small two- and four-colour Ramsey numbers

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Abstract

We give upper bounds for the Ramsey numbers $\mathrm{R}(3,k)$ with $24 \le k \le 50$, for which Radziszowski's survey lists no upper bounds. They come from an exact bootstrap of lower bounds on the number of edges of triangle-free graphs with bounded independence number. It starts from the published values and lower bounds of Goedgebeur and Radziszowski. For instance, $\mathrm{R}(3,24) \le 201$, $\mathrm{R}(3,30) \le 297$ and $\mathrm{R}(3,50) \le 729$. With in addition the table of Backelin's unrefereed report, the bounds are 1 to 3 lower for $24 \le k \le 36$ except $k = 35$.

Deleting vertices from the quartic residue graphs of Lindsay and Cain gives $\mathrm{R}(4,21) \ge 260$ and $\mathrm{R}(4,24) \ge 370$. Finally, we use the four-colour form of the neighbourhood recurrence, with $\mathrm{R}(3,4,4) \le 73$ and $\mathrm{R}(4,4,4) \le 198$ from a companion paper and the survey's $\mathrm{R}(3,3,3,4) \le 149$. This gives $\mathrm{R}(3,3,4,4) \le 442$, $\mathrm{R}(3,4,4,4) \le 1521$ and $\mathrm{R}(4,4,4,4) \le 6082$. All certificates and witnesses are checked exactly by programs in the ancillary files.

1 Introduction

For integers $s,t \ge 1$, the Ramsey number $\mathrm{R}(s,t)$ is the least n such that every red-blue colouring of the edges of K_n contains a red K_s or a blue K_t; equivalently, every graph on n vertices contains a K_s or an independent set of size t. Multicolour numbers $\mathrm{R}(t_1,\dots,t_r)$ are defined in the same way. The current state of knowledge is collected in Radziszowski's dynamic survey [9], whose revision 18 we use for all comparisons, together with the later preprint [12].

This paper is a companion to a study of three-colour numbers [3], in which local Ramsey bounds inside colour neighbourhoods are combined with double counting and with flag-algebra certificates. It gives new bounds for two-colour numbers. The upper bounds are obtained by adapting the ladder procedure of [3, Section 5], and the lower bounds by deleting vertices from quartic residue graphs. It also derives consequences for four-colour numbers. Our results are listed in Table 1.

The upper bounds for $\mathrm{R}(3,k)$ (Section 2) and the lower bounds for $\mathrm{R}(4,k)$ (Section 3) are computer-assisted. Floating-point solvers and local search were used only to find certificates and witnesses. Every certificate and witness is checked in exact arithmetic by a program that does not share code with the search (Section 5).

The four-colour bounds (Section 4) follow by hand from $\mathrm{R}(3,3,3,4) \le 149$ [9] and from $\mathrm{R}(3,4,4) \le 73$ and $\mathrm{R}(4,4,4) \le 198$, which are proved in [3, Theorems 8.3 and 8.2] by

number	previous	new	where
$\mathrm{R}(3,k)$, $24 \le k \le 50$	no upper bound [9]	Table 2	Theorem 2.1
$\mathrm{R}(4,21)$	≥ 242 [11]; ≥ 257 (implicit in [12])	≥ 260	Theorem 3.1
$\mathrm{R}(4,24)$	≥ 343 (implicit in [8])	≥ 370	Theorem 3.1
$\mathrm{R}(3,3,4,4)$	≤ 450 [9]	≤ 442	Theorem 4.1
$\mathrm{R}(3,4,4,4)$	≤ 1576 [9] (implicit in [2])	≤ 1521	Theorem 4.1
$\mathrm{R}(4,4,4,4)$	≤ 6306 [9]; ≤ 6301 (implicit in [2])	≤ 6082	Theorem 4.1

Table 1: Our results. For $\mathrm{R}(3,k)$ with $k \ge 24$, iterating $\mathrm{R}(3,k) \le \mathrm{R}(3,k-1)+k$, which is strict when both terms are even, or Shearer's bound [10], gives weaker explicit bounds (Table 2). Where the two rows of Table 2 differ, the smaller bound also uses the unrefereed report [1].

certificates checked in exact arithmetic by computer.

2 Upper bounds for $\mathrm{R}(3,k)$

A $(3,k;n)$-graph is a triangle-free graph on n vertices with no independent set of size k. A $(3,k;n,e)$-graph is one with e edges, and $e(3,k,n)$ is the least such e. As in [3, Section 3], $m_k(n)$ is a *valid lower bound* if every $(3,k;n)$-graph has at least $m_k(n)$ edges.

For a vertex v of a $(3,k;n,e)$-graph G, $N(v)$ is independent, so $d_v \le k-1$. Moreover, $G_v = G - N[v]$ is a $(3,k-1;n-1-d_v)$-graph, as an independent set of G_v extends by v. As no edge lies inside $N(v)$, G_v has $e - Z_v$ edges, where

$$Z_v = \sum_{u \in N(v)} d_u.$$

As $\sum_v Z_v = \sum_v d_v^2$, valid lower bounds m_{k-1} give

$$\sum_v \left(d_v^2 + m_{k-1}(n-1-d_v) \right) \le ne.$$

This is the inequality of Graver and Yackel [6], on which Goedgebeur and Radziszowski [5] base their bounds for $k \ge 11$. It is the two-colour analogue of the density inequality [3, Lemma 3.3]. As $ne = \sum_v nd_v/2$, it says

$$\sum_v \varphi(d_v) \le 0 \qquad \text{where} \qquad \varphi(d) = d^2 + m_{k-1}(n-1-d) - nd/2.$$

The bootstrap. We bootstrap as in the ladder of [3, Section 5], with this inequality in place of the density inequality. Level $k \ge 11$ uses the bound B_{k-1} on $\mathrm{R}(3,k-1)$ and the valid lower bounds m_{k-1} from level $k-1$. Level 11 starts from $B_{10} = 42$ [5, Theorem 6] and the values and lower bounds for $e(3,10,n)$ in [5].

Level k treats the orders $n = 0, 1, 2, \ldots$ in turn, records a valid lower bound $m_k(n)$, and stops at the first order it excludes, B_k. So $\mathrm{R}(3,k) \le B_k$. At order n the procedure is as follows.

(i) The admissible degrees are the $d \le \min(k-1, n-1)$ with $n-1-d \le B_{k-1}-1$.

(ii) The order n is excluded if $\varphi(d) > 0$ at every admissible d, or if every possible e, up to $\lfloor n(k-1)/2 \rfloor$, is excluded by the extra step below.

(iii) Otherwise $m_k(n)$ is the larger of the bound of [5, Theorem 2] and $\lceil n\alpha/(2\beta) \rceil$, for a line $\alpha - \beta d$ with $\beta > 0$ through an edge of the lower convex hull of the points $(d, \varphi(d))$, d admissible. The line lies below φ at every admissible d, so

$$0 \geq \sum_v \varphi(d_v) \geq n\alpha - 2\beta e.$$

Deleting a vertex as in [3, Lemma 3.5(a)], which gives $m_k(n-1) + \max\{0, n - B_{k-1}\}$, never does better.

An extra step. We add one step. When these bounds give $m_k(n) = e_0$, we enumerate every degree sequence with sum $2e_0$, admissible entries and largest entry at most $e_0 - m_k(n-1)$ (as $G - v$ is a $(3, k; n-1)$-graph) that satisfies the inequality with $e = e_0$.

We then try to exclude each one by a small linear programme in the numbers of edges inside and between degree classes. Its constraints hold for every $(3, k; n, e_0)$-graph with that degree sequence:

- the edge ends at the class of degree a number a times the class size;
- the sum of Z_v over the class of degree a is at most the class size times $e_0 - m_{k-1}(n-1-a)$;
- it is at least the class size times $e_0 - \lfloor (n-1-a)(k-2)/2 \rfloor$, as G_v has maximum degree at most $k-2$;
- the number of edges between two classes is at most the product of their sizes;
- Mantel's bound holds on every union of classes;
- a vertex of degree a has at least t neighbours of degree at most b, so the class of degree a sends at least t times its size edge ends to classes of degree at most b. Here t is least such that the t smallest entries at most b, other than that of the vertex itself, together with $a - t$ times the least entry above b, sum to at most $e_0 - m_{k-1}(n-1-a)$. This sum is a lower bound for Z_v when v has exactly t such neighbours, and it decreases as t grows.

If every sequence is excluded, so is e_0, and the step is repeated with $e_0 + 1$. Each exclusion is certified either by the failure of the neighbour-count test, when for some class no such t exists, or by rational Farkas multipliers for named constraints.

Theorem 2.1 (computer-assisted)**.** *From the published values and lower bounds of [5, Theorem 2 and Tables 3–5] and* $\mathrm{R}(3, 10) \leq 42$ *[5, Theorem 6],* $\mathrm{R}(3, k)$ *is at most the value in the row "[5] only" of Table 2 for* $24 \leq k \leq 50$*. With in addition the lower bounds on* $e(3, k, n)$ *in the table of [1, Section 5], it is at most the value in the row "with [1]".*

The two chains. Above level 10, each chain uses only the bounds B_{k-1} that it proves itself, never the unpublished values of [9]. Neither chain needs the bound $\mathrm{R}(3, 10) \leq 41$ listed in [9]. The report [1] is a preprint that gives little of its proofs, so the second row rests on inputs without published proofs. The first row is 1 to 3 larger for $24 \leq k \leq 34$ and $k = 36$, and equals the second for $k = 35$ and $k \geq 37$.

k	24	25	26	27	28	29	30	31	32
[5] only	201	216	231	247	264	280	297	315	333
with [1]	200	215	230	245	261	278	295	312	331
k	33	34	35	36	37	38	39	40	41
[5] only	351	370	389	409	429	450	471	492	513
with [1]	349	369	389	408	429	450	471	492	513
k	42	43	44	45	46	47	48	49	50
both	536	559	582	606	630	654	679	704	729

Table 2: Upper bounds for $\mathrm{R}(3,k)$ from Theorem 2.1. The two rows coincide for $k = 35$ and $k \geq 37$, and for $k \geq 42$ they are shown as one, "both". The survey [9] lists no upper bounds for these k. For comparison, $\mathrm{R}(3,k) \leq \mathrm{R}(3,k-1) + k$, strict when both terms are even, iterated from the survey's $\mathrm{R}(3,23) \leq 185$, gives 209, 234, 259, 371 and 1171 for $k = 24, 25, 26, 30, 50$. Shearer's bound [10] $\mathrm{R}(3,k) \leq \lfloor (k-1)/f(k-1) \rfloor + 1$, with $f(d) = (d \ln d - d + 1)/(d-1)^2$, gives 223 for $k = 24$ and 792 for $k = 50$.

Without the step above, the chain from [5] alone gives the same bounds for $11 \leq k \leq 50$, except $\mathrm{R}(3,20) \leq 146$ and $\mathrm{R}(3,35) \leq 390$. The chain from [5] alone also raises seven lower bounds of [5, Tables 9–11] by one, to

$$e(3,13,58) \geq 261, \qquad e(3,14,69) \geq 366, \qquad e(3,14,70) \geq 382,$$
$$e(3,15,n) \geq 516,\ 534,\ 553 \text{ and } 573 \quad \text{for } n = 82,\ldots,85.$$

Backelin's unpublished tables may already contain these.

Checking the certificates. The certificates are checked by a program that does not import the generating code. It enumerates the degree sequences itself, rebuilds each named constraint from its definition, and verifies the multipliers in exact arithmetic.

- The chain with [1] has one row for each level $11 \leq k \leq 50$ and each order $0 \leq n < B_k$, 13,376 rows in all. It excludes 765 degree sequences at 53 triples (k,n,e), 686 by the neighbour-count test and 79 by Farkas multipliers.
- The chain from [5] alone has 13,403 rows and excludes 766 sequences at 50 triples, 684 and 82 respectively.

Comparison with the survey. For $k \leq 23$ the procedure improves no value of [9]. From [5] alone it gives $\mathrm{R}(3,17) \leq 109$ and $\mathrm{R}(3,20) \leq 145$, which [9] credits to an unpublished manuscript of Backelin and to Lesser's thesis [7], respectively. Adding the two values $e(3,11,39) \geq 119$ and $e(3,12,43) \geq 129$ of [1] lowers its $\mathrm{R}(3,16) \leq 98$ and $\mathrm{R}(3,19) \leq 133$ to 97 and 132, as stated in [1].

With the whole table of [1] it gives $\mathrm{R}(3,23) \leq 185$, against 187 from [5] alone. The survey [9] credits this bound to a personal communication of Backelin; it now has a derivation that can be checked, given the inputs of [1]. At $k = 18$, 21 and 22 this chain gives 121, 158 and 172, one more than the unpublished values of [9] (from [5] alone, 121, 159 and 173).

For $k = 18$, at $n = 120$ the hull already gives $e \geq 1020 = 120 \cdot 17/2$, so only the 17-regular

sequence remains. In it every G_v would be a $(3, 17; 102, 731)$-graph. So $e(3, 17, 102) \geq 732$, one more than our bound, would give $\mathrm{R}(3, 18) \leq 120$.

3 Lower bounds for $\mathrm{R}(4, k)$

For a prime $p \equiv 1 \pmod 8$ let Q_p be the circulant graph on $\mathbb{Z}_p$ in which $x \sim y$ if and only if $x - y$ is a nonzero fourth power. The relation is symmetric, as -1 is a fourth power. Lindsay and Cain [8] showed that Q_{313} and Q_{457} are K_4-free with no independent sets of size 22 and 25, respectively. Their independence numbers are

$$\alpha(Q_{313}) = 21 \qquad \text{and} \qquad \alpha(Q_{457}) = 24.$$

If a vertex set S meets every independent set of size $\alpha(Q_p)$, then $Q_p - S$ is K_4-free and has independence number less than $\alpha(Q_p)$.

Theorem 3.1 (witnesses checked by computer)**.** $\mathrm{R}(4, 21) \geq 260$ *and* $\mathrm{R}(4, 24) \geq 370$.

Proof. Step 1: symmetries and cosets. The maps $x \mapsto ax + b$ with a a nonzero fourth power are automorphisms of Q_p. Let O_1, O_2, O_3 be the cosets of the group of nonzero fourth powers in $\mathbb{Z}_p^*$ other than the group itself, in a fixed order, and let r_j be the least element of O_j.

Step 2: representatives. Let $k \geq 2$, let J be an independent k-set and let $u \in J$. The translate $J - u = \{x - u : x \in J\}$ is an independent k-set that contains 0, so its other elements are not fourth powers and lie in $O_1 \cup O_2 \cup O_3$. Let j be least with $J - u$ meeting O_j, say at w. Then r_j/w is a fourth power, and multiplying by it maps $J - u$ to an independent k-set that contains 0 and r_j and still avoids $O_1, \ldots, O_{j-1}$, as multiplication by a fourth power fixes each coset. Call such a set a representative. So every independent k-set is the image of a representative under one of the maps above.

Step 3: the computation. An exhaustive search lists all representatives for $(p, k) = (313, 21)$ and $(457, 24)$. A weighted local search found sets S of 54 and 88 vertices, listed in the ancillary files, and an exact program checks that each meets every image of every representative. So $Q_{313} - S$ is a K_4-free graph on 259 vertices with no independent 21-set, and $Q_{457} - S$ one on 369 vertices with no independent 24-set. □

Verification. Both graphs are rebuilt from the lists of deleted vertices by programs written independently of the search. The 259-vertex graph is also checked by a direct exhaustive search for K_4 and for independent 21-sets, and the 369-vertex graph by one for K_4 and for independent 24-sets. Neither search uses symmetry, so each is independent of the orbit argument above. □

Comparison with earlier bounds. The previous bounds are as follows.

- For $\mathrm{R}(4, 21)$, the survey has 242 [11], and $\mathrm{R}(4, 20) \geq 252$ from a recent preprint [12] gives 257 by [9, 2.3(e)].
- For $\mathrm{R}(4, 24)$, the non-neighbourhood of a vertex of Q_{457} (Q_{457} is 114-regular) has 342 vertices, is K_4-free and has no independent 24-set, as such a set would extend by the

vertex. This gives $\mathrm{R}(4,24) \geq 343$. The survey lists no value for $\mathrm{R}(4,24)$. Its rules give 324, by applying [9, 2.3(e)] twice to $\mathrm{R}(4,22) \geq 314$ [8].

The method does not reach $\mathrm{R}(4,23) \geq 320$, which would improve the bound 319 that [9, 2.3(e)] gives from $\mathrm{R}(4,22) \geq 314$. That needs a set of at most $457 - 319 = 138$ vertices meeting every independent 23-set of Q_{457}, and the smallest we found has 147.

4 Four colours

For $\mathbf{t} = (t_1, \ldots, t_4)$ and the unit vectors $\mathbf{e}_i$, the four-colour form of the neighbourhood recurrence [9, 6.1(a)] is

$$\mathrm{R}(\mathbf{t}) \leq \sum_{i=1}^{4} \mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 2.$$

Indeed, consider a colouring of K_n with no colour-j K_{t_j} for any j. The colour-i neighbourhood of a vertex v contains no colour-i $K_{t_i - 1}$, which would extend by v, and no colour-j K_{t_j} for $j \neq i$. So it has at most $\mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 1$ vertices. Summing over i gives

$$n - 1 \leq \sum_{i} (\mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 1).$$

The inequality is strict when its right-hand side is even and some term $\mathrm{R}(\mathbf{t} - \mathbf{e}_i)$ is even. At the order one less, which is odd, the count is then tight. So every vertex has colour-i degree exactly $\mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 1$, which is odd, contradicting the handshake lemma.

Both statements hold with the same proofs when each $\mathrm{R}(\mathbf{t} - \mathbf{e}_i)$ is replaced by a proven upper bound for it. An entry equal to 2 may be dropped, leaving a number with fewer colours, as a colouring with no colour-i K_2 has no colour-i edges.

Theorem 4.1. $\mathrm{R}(3,3,4,4) \leq 442$, $\mathrm{R}(3,4,4,4) \leq 1521$ *and* $\mathrm{R}(4,4,4,4) \leq 6082$.

Proof. With $\mathrm{R}(3,4,4) \leq 73$ [3, Theorem 8.3] and $\mathrm{R}(3,3,3,4) \leq 149$ [9], the recurrence gives

$$\mathrm{R}(3,3,4,4) \leq 2 \cdot 73 + 2 \cdot 149 - 2 = 442.$$

Parity does not apply, as every term is odd.

With $\mathrm{R}(4,4,4) \leq 198$ [3, Theorem 8.2] it then gives

$$\mathrm{R}(3,4,4,4) \leq 198 + 3 \cdot 442 - 2 = 1522.$$

This is even, as are all four terms, so $\mathrm{R}(3,4,4,4) \leq 1521$.

Finally,

$$\mathrm{R}(4,4,4,4) \leq 4 \cdot 1521 - 2 = 6082,$$

and parity does not apply, as 1521 is odd. □

The inputs and earlier bounds. The two inputs from [3] are computer-assisted. The survey's $R(3,3,3,4) \le 149$ is itself the recurrence with parity, $3 \cdot 30 + 62 - 2 = 150$, from $R(3,3,4) = 30$ and $R(3,3,3,3) \le 62$ [4].

With the survey's $R(3,3,4,4) \le 450$ in place of 442, that is, without $R(3,4,4) \le 73$, the bound $R(4,4,4) \le 198$ alone gives $R(3,4,4,4) \le 1545$ and $R(4,4,4,4) \le 6178$. We have not tried the analogue of the parity clause modulo 3 of [2], which gives the survey's $R(3,4,4,4) \le 1576$.

Four triangle-free colours. We also tried to improve $R(3,3,3,3) \le 62$ [4], that is, to exclude colourings of K_{61} with four triangle-free colours, by certificates analogous to those for $R(4,4,4)$ in [3, Sections 7 and 8]. Such a certificate exists only if a flag relaxation, here with four- or five-vertex patterns and with four-colour analogues of the local caps of [3, Lemmas 2.1 and 2.3] among its constraints, is infeasible.

At $n = 61$ every colour degree is at most $R(3,3,3) - 1 = 16$ and the four sum to 60. So the *profile* of a vertex, its sorted quadruple of colour degrees, is one of

$$(12,16,16,16), \quad (13,15,16,16), \quad (14,14,16,16), \quad (14,15,15,16), \quad (15,15,15,15).$$

Both relaxations are numerically feasible. Floating-point solvers found an assignment of pattern densities meeting all their constraints, even when every vertex is required to have profile $(12,16,16,16)$. The four-vertex relaxation was solved as a linear programme by HiGHS, with its semidefinite constraint imposed by eigenvector cuts until no eigenvalue was below -10^{-8}. The five-vertex relaxation was solved as a semidefinite programme by Clarabel at tolerance 10^{-8}. So we obtained no such certificate from these relaxations, and no bound follows.

5 Verification, literature and limitations

Ancillary files. The ancillary directory, the same as for [3], contains the programs and data:

- `two_colour_r3k/` for Theorem 2.1, with the inputs from [5] and [1] and the certificates of both chains;
- `two_colour_r4k/` for Theorem 3.1, with the lists of deleted vertices, the adjacency matrices of the two graphs and the logs of the direct searches.

The two inputs of Theorem 4.1 from [3] are checked by the programs of that paper in the same directory.

The checkers are written in Python and C with their standard libraries. The C programs use POSIX threads. Negative controls (a weakened input, a removed multiplier or sequence certificate, a value raised by one, a forged cap, a deleted vertex put back) are all rejected.

Literature. We searched arXiv and the web up to 23 October 2026 for earlier statements of our new bounds, and found none.

- For $R(3,k)$ with $k \ge 24$ the survey lists no upper bounds, and the upper bounds of [5] and of Lesser's thesis [7] stop at $k = 16$ and $k = 21$. Backelin's unpublished manuscript, to which [9] credits $R(3,17) \le 109$, may contain bounds of this kind.

- For $\mathrm{R}(4,21)$ the best previous bound we know of, 257, follows from $\mathrm{R}(4,20) \geq 252$ in the preprint [12], which we have not checked. Without it the bound is 242 [11].
- For $\mathrm{R}(4,24)$ the survey lists nothing, and 343 is implicit in [8].
- For the four-colour numbers the previous bounds are those of [9] and [2] in Table 1.

Such a search cannot establish priority, and we would welcome pointers to any earlier source.

Limitations. For $k \leq 23$ the bootstrap of Section 2 improves none of the bounds in [9]. At $k = 18$ it would give $\mathrm{R}(3,18) \leq 120$ if its own lower bound $e(3,17,102) \geq 731$ were one larger. The second row of Table 2 rests on the lower bounds of [1], whose proofs are largely unpublished.

The deletion method of Section 3 does not reach $\mathrm{R}(4,23) \geq 320$. The four-colour bounds rest on $\mathrm{R}(3,4,4) \leq 73$ and $\mathrm{R}(4,4,4) \leq 198$ [3, Theorems 8.3 and 8.2], which are computer-assisted, and on $\mathrm{R}(3,3,3,4) \leq 149$, which [9, 6.1(a)] derives from $\mathrm{R}(3,3,4) = 30$ and $\mathrm{R}(3,3,3,3) \leq 62$ [4], both computer-assisted.

Acknowledgements

Much of the work reported here was carried out with Claude Opus 5.5, a large language model developed by Anthropic, used under the direction of the author. It searched for the arguments and certificates, wrote the search and checking programs in the ancillary files, and drafted the text. Every certificate and witness is checked in exact arithmetic by the programs in the ancillary files, which can be run independently of the model. The author takes full responsibility for the contents of this paper.

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